

Chapter-VI

Junction Field Effect Transistor

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Introduction to J-FET, Types of J-FET, V-I Characteristics of J-FET in CS configuration, FET as an amplifier, J-FET Biasing.

6.0 OBJECTIVES

After completion of this chapter we should be able to understand :

- To know the principle of operation of JFET and MOSFET.
- To learn the analysis of field effect transistor using parameters.
- To know the various biasing methods on field effect transistor.
- Characteristics of field effect transistors in different configurations.

6.1 INTRODUCTION

The Field-effect transistor amplifiers provide an excellent voltage gain with an added feature of high input impedance. They are also low-power-consumption configurations with good frequency range and minimal size and weight.

The FET device controls an output (drain) current by means of a small input (gate-voltage) voltage. FET is a voltage-controlled device. In both cases, however, note that the output current is the controlled variable. Because of the high input characteristic of FETs, the ac equivalent model is somewhat simpler than that employed for BJTs. The FET can be used as a linear amplifier or as a digital device in logic circuits. FET devices are also widely used in high-frequency applications and in buffering (interfacing) applications.

6.2 FET (FIELD EFFECT TRANSISTOR)

Field Effect Transistor: This is a semiconductor device in which the output current is controlled by the applied electric field. The current in a FET is carried only by one type of majority charge carriers-electrons (or) holes. It is also called as *unipolar transistor*.

The FETs are unipolar devices because they operate only with one type of charge carrier. If the channel conduction and hence the current in the device from source to drain can be controlled by the application of an electric potential or transverse electric field introduced by gate, the device is known as Field effect transistor (FET).

The term field effect stands for the fact that the operation of the device mainly depends on the electric field applied between its terminals called Gate and source analogous to the base and collector in transistor to control the current through it. The major feature of FETs is their high input resistance.

The field effect transistor is available in several forms. Two most important forms are,

1. Junction field effect transistor (JFET)
2. Insulated gate field effect transistor (IGFET) or metal oxide semiconductor field effect transistor (MOSFET) or (MOST).

6.2.1 Comparison of FET over BJT

FET	BJT
The relationship between input and output quantities in non-linear due to square term in shockley's equation is, $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$	The BJT is an almost linear device or BJT works linearly in active region as an amplifier.
FET's are named as source, drain and gate.	The BJT is three terminals can be named as the base, emitter and collector.
The Higher frequency response.	Frequency variation affects the performance.
FET is an Unipolar semiconductor device because of its operation can be depends upon the flow of majority carriers i.e., either holes or electrons as the case may be.	The BJT is a bipolar semiconductor device because the current constituting elements are both majority carriers.
The input impedance of FET is much more larger than BJT. The reason behind this is that an input terminal i.e., gate to source of FET can be reverse biased and reverse bias offers ideally infinite resistance.	The input impedance of BJT is very less in comparison to FET.
It is costlier than BJT	It is relatively cheaper
It have good thermal stability because of absence of minority carriers.	These are temperature dependent, thermal runaway may cause.
FET is a voltage controlled device	BJT is a current controlled device
Small gain bandwidth product	Greater than FET
Small sized	Comparatively bigger

6.2.2 Classification of FET

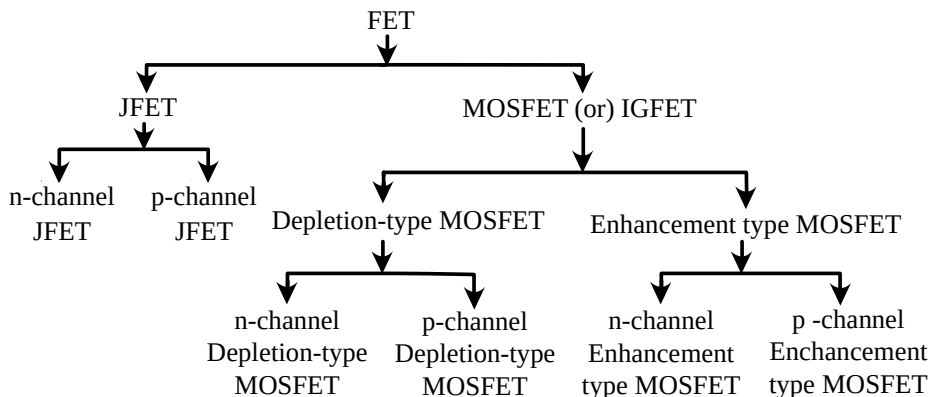


Figure 6.1: Different types of field effect transistors

6.3 JFET

Junction Field Effect Transistor (JUGFET or JFET) has no PN-junctions but instead has a narrow piece of high-resistivity semiconductor material forming the “Channel” of either N-type or P-type silicon for the majority carriers to flow through with two ohmic electrical connections at either end commonly is called the Drain and Source respectively.

Based on the majority carriers, JFET has been classified into two types, namely,

1. N-Channel JFET with electrons as the majority carriers, and
2. P-Channel JFET with holes as the majority carriers.

Likewise, the P-channel JFET’s channel can be doped with acceptor impurities meaning that the flow of current through the channel is positive (hence the term P-channel) in the form of holes. N-channel JFET’s have a greater channel conductivity (lower resistance) than their equivalent P-channel types. Since electrons have a higher mobility through a conductor compared to holes. This makes the N-channel JFET’s a more efficient conductor compared to their P-channel counterparts.

Symbols: JFET is a unipolar-transistor, which acts as a voltage controlled current device and is a device in which current at two electrodes is controlled by the action of an electric field at a p-n junction. It consists of three terminals like Gate (G), Drain (D) and Source (S).

In these 3 terminals, the Gate terminal acts as a controlling terminal. BJT acts as a current controlling device. FET also acts as a voltage controlling device. Here, the voltage between gate and source controls the drain current. Hence, it is called as voltage controlled device.

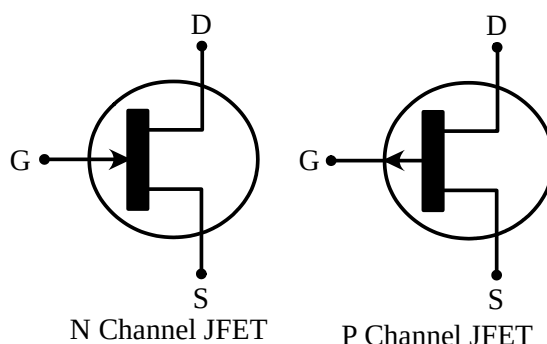


Figure 6.2: Symbols of JFET

6.3.1 Construction of JFET

The junction field effect transistor is a three terminal semiconductor device in which current conduction is by one type of carrier i.e., electrons and holes. JFET consists of p-type or n-type silicon material containing two pn junction at the sides.

If the bar is n-type is called as n-channel JFET and when the bar is p-type is called as p-channel JFET. The two pn junctions forming diodes can be connected internally and common terminal is called as gate is taken out. Other terminals are source and drain.

P-channel JFET: P-channel JFET consists of p-type silicon bar forming a conducting channel for the charge carriers. The heavily doped n-regions introduced on both sides of bar form the gate. It can be used to control the flow of holes from source to drain. Construction of JFET as shown in figure 6.3.

N-channel JFET: N-channel JFET consists of n-type silicon bar forming the conducting channel for the charge carriers. The heavily doped p-regions introduced on both sides of the bar form the gate. It can be used to control the flow of electrons from source to drain.

- **Source:** This terminal is connected to the negative pole of the battery. Electrons which are the majority carriers in the N- type bar enter the bar through this terminal.
- **Drain:** This terminal is connected to the positive pole of the battery. The majority carriers leave the bar through this terminal.
- **Gate:** Heavily doped p- type silicon is diffused on both sides of the N- type silicon bar by which PN- junction are formed. These layers, are joined together and called gate G..
- **Channel:** The region BC of the N- type bar between the depletion region is called the channel. Majority carriers move from the source to drain when a potential difference V_{DS} is applied b/w the source & drain.

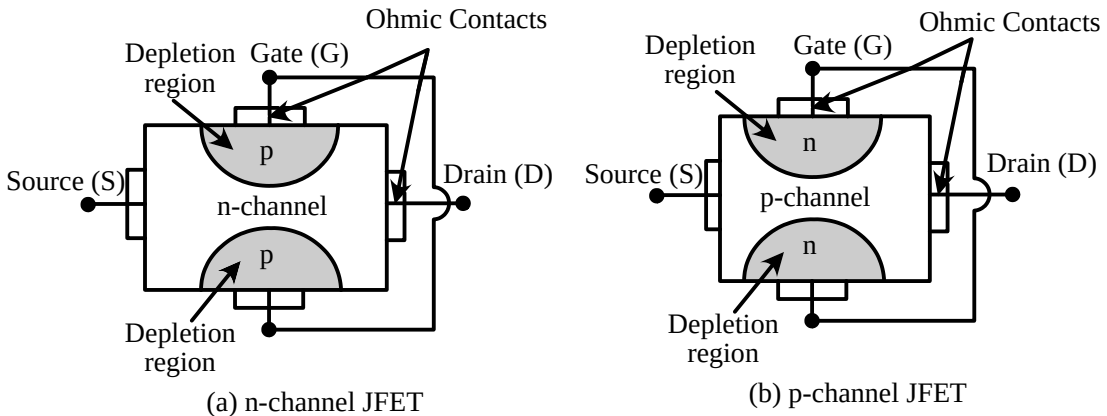


Figure 6.3: JFET Construction

6.3.2 JFET Polarities

Figure 6.4 (a) shows the n-channel JFET polarities whereas figure 6.4 (b) shows p-channel JFET polarities. In each case, the voltage between the gate and source is such that the gate can be reverse biased. This is the normal way of JFET connection. The drain and sources are interchangeable i.e., either end can be used as source and an other end as drain.

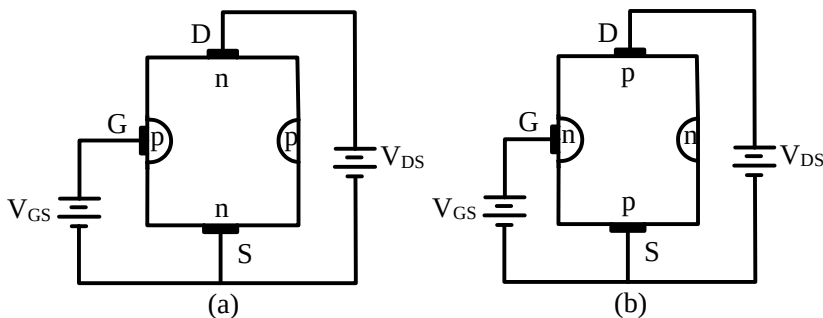


Figure 6.4: JFET polarities

6.3.3 Operation and Working of JFET

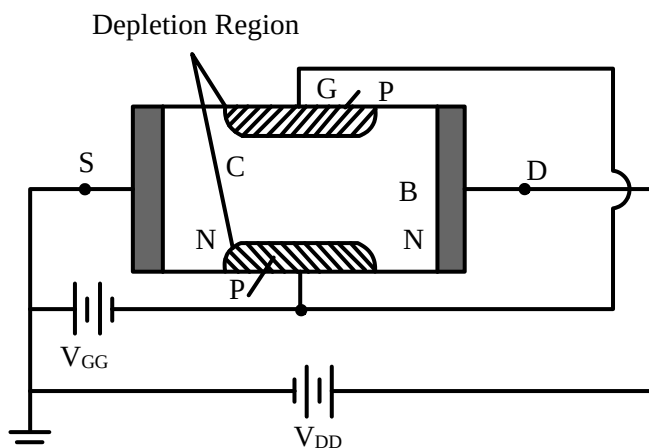


Figure 6.5: JFET operation

To consider the operation of a JFET as shown in figure 6.6. The application of voltage $V_{DS} = V_{DD}$ drain supply voltage from drain to source will cause the electrons to flow through the channel. The amount of drain current I_D can be initially determined by the value of V_{DS} since it is just the ohmic resistance of bar from S to D i.e., limiting the current.

In order to see what effect will be a voltage V_{GS} from gate to source, have on this drain current for a given drain to source voltage, if the P-N junctions between gate and source are under reverse-bias conditions. These reverse biased P-N junctions can develop depletion regions as shown by cross-hatching in figure 6.4.

The depletion regions are non-conductive except for a few intrinsic carriers and decrease the conductance of channel between source and drain electrodes. If the reverse bias is increased, the size of the depletion regions increases and drain current can be reduced.

If the reverse current is large enough for two depletion regions to meet the channel becomes pinched off and drain current cuts-off. The reverse bias can be required for pinch off is labelled as V_p . The drain current through the channel depends upon the degree to which the electric field (due to the change in the depletion region) extends into channel and provides the effect of decreasing the conductance through an transistor. Thus, the name field-effect transistor (FET) given to this device.

In a bar of P-type semiconductor, the gate is formed due to N-type semiconductor. Working of the P-channel JFET will be similar to that of N-channel JFET with the proper alterations in the biasing circuits. In this case holes will be the current carriers instead of electrons.

The circuit symbols for N-channel and P-channel JFETs are shown in figure 6.6. It should be noted that the direction of arrow points in the direction of conventional current which would flow into gate if the PN junction was forward biased.

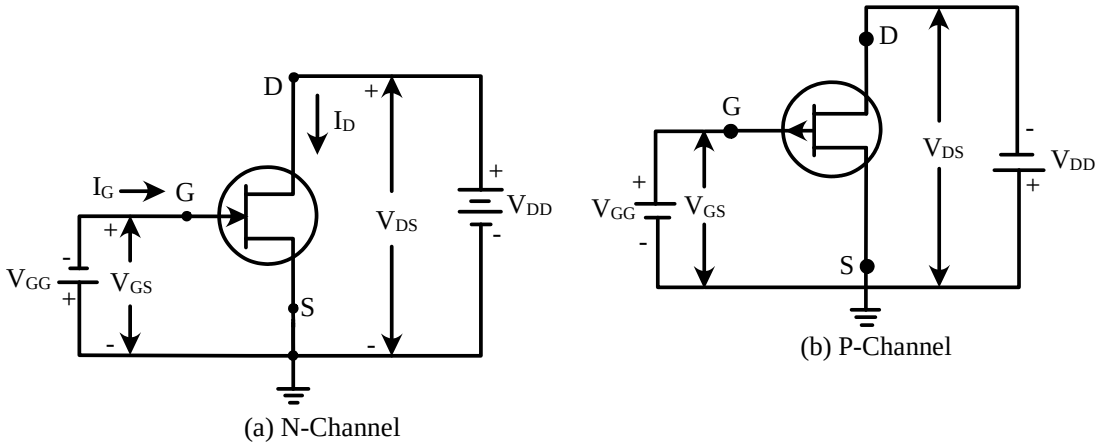


Figure 6.6: N-channel JFET

6.3.4 Drain Current

To find an expression for drain current, the current in the channel resistance is proportional to V_{DS} in accordance with the Ohm's law, and to $V_{GS} - V_P$ because of pinch off. The finite drain voltage also results in an additional reverse bias varying from zero at source end to V_{DS} at the drain end.

The average reverse bias can be taken as $\frac{V_{DS}}{2}$. The drain current is,

$$I_D = K \left[V_{GS} - V_P - \left(\frac{V_{DS}}{2} \right) \right] V_{DS}$$

Where K is the constant.

At pinch off, the sum of gate voltage and drain voltage is $V_{GS} + V_{DS} = V_P$, so that the above equation becomes,

$$I_D = K \left[V_{GS} - V_P - \left(\frac{V_{GS} - V_P}{2} \right) \right] (V_{GS} - V_P) = \frac{K}{2} (V_{GS} - V_P)^2$$

6.4 WORKING OPERATION OF N-CHANNEL JFET

Case (i): When $V_{GS} = 0$ and $V_{DS} = 0$

If no voltage is applied between drain and source, gate and source, the thickness of depletion regions round the P-N junction is uniform as shown in figure 6.7. Here, the drain current $I_D = 0$ if $V_{DS} = 0$. The depletion region around the P_D junctions are of equal thickness and symmetrical.

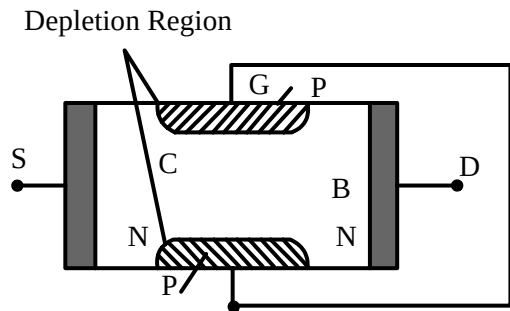


Figure 6.7: Thickness of the depletion regions

Case (ii): When $V_{DS} = 0$ and V_{GS} is decreased

The P-N junctions are reverse-biased and hence the thickness of depletion region can be increased. If V_{GS} is decreased from zero, the gate reverse bias increases. Thickness of the depletion region in the channel increases until the two depletion regions make contact with each other. In this condition, the channel is said to be cut-off and gain current becomes zero. Value of V_{GS} which is required to cut-off the channel is called cut-off voltage V_C .

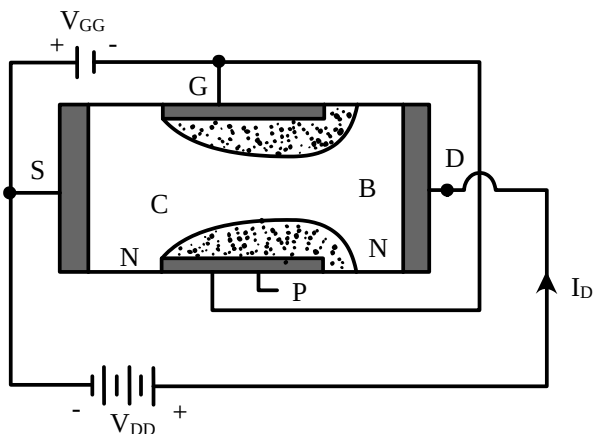


Figure 6.8: JFET under Applied Bias

Case (iii): When $V_{GS} = 0$ and V_{DS} is increased from Zero

In this case the drain is positive with respect to the source. Now the majority carriers (electrons) flow through the N-Channel from source to drain. Thus, the conventional current I_D flows from drain to source. The magnitude of the current can depend upon the following factors:

1. Conductivity of channel
2. Length of channel
3. Cross-sectional area 'A' of channel.
4. Magnitude of applied voltage V_{DS} .

The channel acts as a resistor of resistance 'R' is,

$$I_D = \frac{V_{DS}}{R} = \frac{AV_{DS}}{\rho L}$$

Where ' ρ ' is the resistivity of channel.

6.5 DRAIN AND TRANSFER CHARACTERISTICS OF JFET

Drain Characteristics of JFET: If V_{DS} increases, the reverse voltage across the P-N junction increases and hence the thickness of depletion region can also increase. Hence, the channel is wedge shaped as shown in figure 6.8. If V_{DS} increases at a certain value V_p of V_{DS} , the cross-sectional area of channel becomes minimum. At this voltage, an channel is said to be pinched off and drain voltage V_p is called as pinch-off voltage. Decreasing cross-section of channel with an increase of V_{DS} then following results can be obtained.

- a) If V_{DS} is increased from zero, I_D increases linearly along OP, this region from $V_{DS} = 0$ to $V_{DS} = V_p$ is called as an ohmic region. In this region, the FET acts as a voltage variable resistor (VVR) or the voltage dependent resistor (VDR).

- b) If $V_{DS} = V_p$, I_D becomes maximum.

When V_{DS} increased beyond V_p , the length of pinch-off (or) saturation region increases. Thus, there is no further increase of I_D .

- c) At a certain voltage corresponding to the point B, I_D suddenly increases. Due to this effect is the Avalanche multiplication of electrons can be caused by breaking of the covalent bonds of silicon atoms in the depletion region between gate and drain. Drain voltage at which the breakdown can be occurs is denoted by BV_{DGO} .

- iv) **When V_{GS} is negative and V_{DS} is increased:** If the gate can be maintained at a negative voltage less than negative cut-off voltage, the reverse

voltage across the junction is further increased. Since for a negative value of V_{GS} , the curve of I_D versus V_{DS} is similar to that for $V_{GS} = 0$, but the values of V_p and BV_{DGO} are lower.

Drain current I_D can be controlled by the electric field that extends into the channel due to reverse biased voltage applied to gate. Thus, this device has been given the name Field Effect Transistor.

Region of Operation: $V_{DS} < V_p$

- i. In this region $I_D \propto V_{DS}$ that is the property of a resistance. So this region in which JFET behaves as a resistance and follows the ohms law called as ohmic region.
- ii. When $V_{DS} > V_p$, it acts as constant current source and it is also known as linear or saturation region.
- iii. At $V_{GS} > -V_p$ ideally $I_D = 0$, devices turns off and it is called as cut-off region.
- iv. At very high value of V_{DS} , breakdown occurs. This maximum value of V_{DSmax} can be specified on the FET data -sheet.

Transfer Characteristics of N-type JFET

To explain the basic operation of N-type JFET we have drawn the graph between drain current I_D and drain to source voltage V_{DS} taking gate to source voltage V_{GS} as a constant. It is called as drain characteristics or output characteristics. With the help of drain characteristics, we can

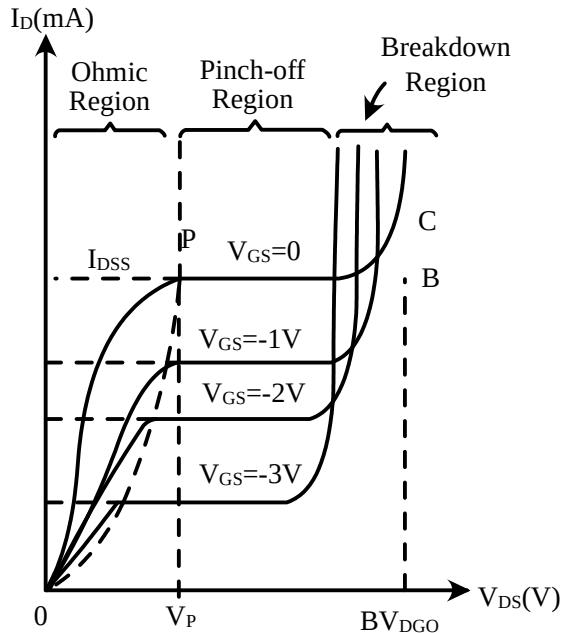


Figure 6.9: Characteristics of n-channel JFET

also find the characteristics showing the relation between drain current and gate to source (V_{GS}) voltage taking (V_{DS}) as constant. It is called transfer characteristics of JFET. A graph of the drain current as a function of the gate voltage (V_{GS}) is called the transconductance curve.

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \Rightarrow \left(1 - \frac{V_{GS}}{V_P} \right) = \sqrt{\frac{I_D}{I_{DSS}}} = \frac{I_D}{I_{DSS}}$$

$$1 - \frac{V_{GS}}{V_P} = \sqrt{\frac{I_D}{I_{DSS}}} \Rightarrow V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right)$$

By reading the values of I_D and V_{GS} from figure 6.10 of depletion region width and p channel FET and its characteristics one can plot the transconductance curve.

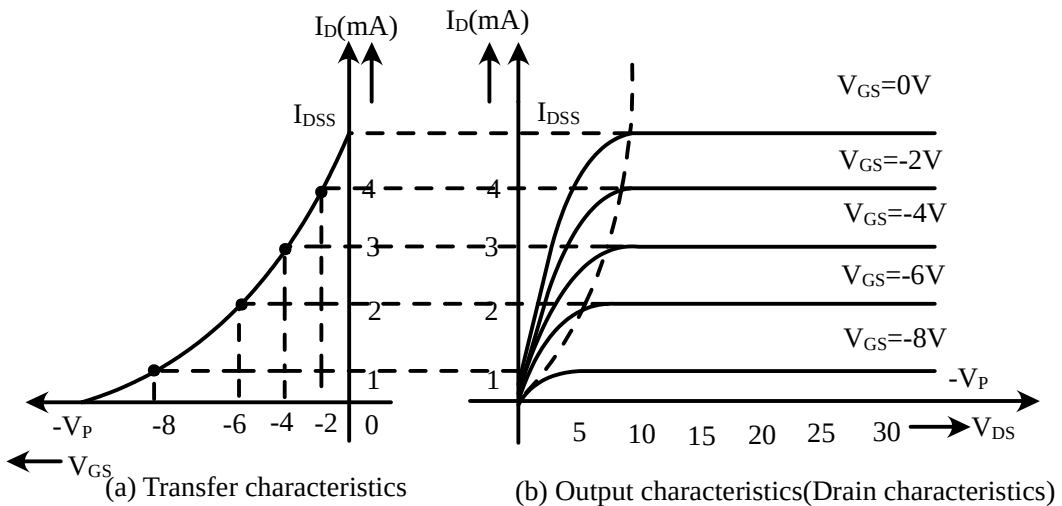


Figure 6.10: Characteristics of N-JFET

Here I_D is taken along y- axis & V_{GS} is taken along x - axis. When $V_{GS} = 0V$ the drain current is maximum (I_{DSS}) when V_{GS} is increased (-ve side) then width of the depletion region increases. Therefore, the drain current decreases. At one particular voltage (V_{GS}), the channel width is said to be completely reduced and that voltage is called as pinch-off voltage. Here the drain current becomes equal to zero.

6.6 P-TYPE JFET

Only difference between P-type and N-type JFET is that P regions can be replaced by N-type and N regions can be replaced by P-type. Because of this, the polarities of voltages are changed.

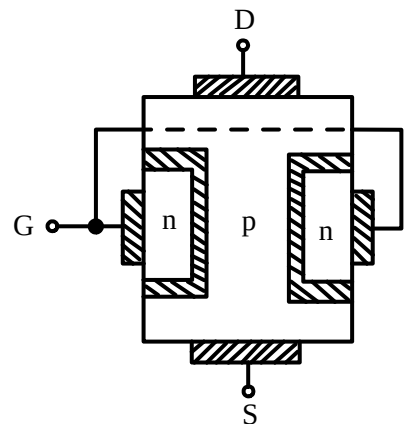


Figure 6.11: P-JFET

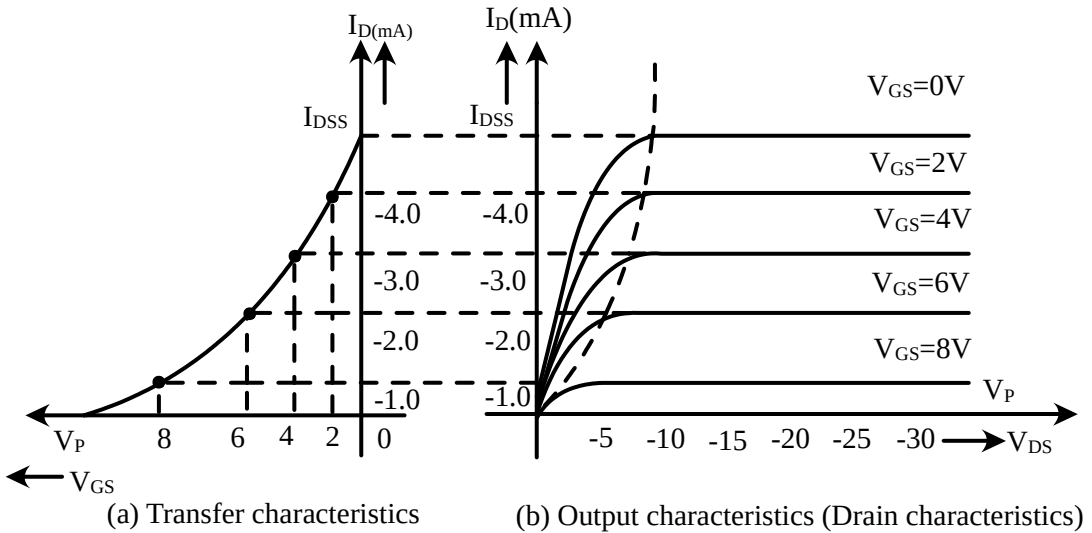


Figure 6.12: Characteristics of P-JFET

Non-linear relation was proved by Shockley's equation. The Shockley equation is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2 \quad \dots (1)$$

Where I_{DSS} and V_p are constants for a particular device

V_p is the pinchoff voltage

The pinch-off voltage V_p is negative for the n-channel JFET and gate-to-source voltage is also negative. Therefore, the ratio $\frac{V_{GS}}{V_p}$ is positive.

Similarly, the pinch-off voltage for the p-channel JFET is positive and gate-to-source voltage V_{GS} is also positive. Thus, the ratio $\left(\frac{V_{GS}}{V_p} \right)$ is always positive.

For the n-channel device, the saturation region occurs when $V_{DS} \geq V_{DS(sat)}$

$$\text{Where } V_{DS(sat)} = V_{GS} - V_p \quad \dots (2)$$

$$\text{For the p-channel device, the saturation region occurs when } V_{SD} = V_p - V_{GS} \quad \dots (3)$$

6.7 DETERMINATION OF FET PARAMETERS

In a JFET, the drain current I_D depends upon the drain voltage V_{DS} and the gate voltage V_{GS} . Any one of these variables may be fixed and the relation between the other two is determined. These relations are determined by the three parameters which defined below.

1. Mutual Conductance (or) Transconductance g_m : It is the slope of the transfer characteristic curves. It can be defined as the ratio of change in drain current to the change in gate to source

voltage at constant drain-source voltage i.e.,

$$g_m = \frac{\text{Change in } I_D}{\text{Change in } V_{GS}}$$

When V_{DS} is constant

$$g_m = \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}} = \frac{\Delta I_D}{\Delta V_{GS}} \mid V_{DS} \text{ is held constant} \quad \dots (1)$$

Here it simplifies the slope of transfer characteristics. Its unit is siemens (S) which was earlier called mho or A/V. It is also called transconductance (g_{fs}) or forward transmittance (Y_{fs}). The values range from 0.1 to 20mS for both JFETs and MOSFETs.

Shockley Equation: Mathematical Expression for g_m

The Shockley equation is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Where I_{DSS} = Drain saturation current

$$I_D = I_{DSS} \text{ when } V_{GS} = 0$$

$$V_P = \text{Pinch-off voltage}$$

Differentiating both sides with respect to V_{GS} . We get,

$$\begin{aligned} \frac{dI_D}{dV_{GS}} &= 2I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right) \left(-\frac{1}{V_P} \right) \\ g_m &= -\frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P} \right) \quad \dots (2) \end{aligned}$$

When $V_{GS} = 0$, $g_m = g_{m_0}$

$$g_{m_0} = -\frac{2I_{DSS}}{V_P} \quad \dots (3)$$

From equation (2) and (3). We get,

$$g_m = g_{m_0} \left(1 - \frac{V_{GS}}{V_P} \right)$$

2. A.C. Drain Resistance (r_d): This is the a.c. resistance between drain and source terminals when JFET operates in the pinch-off region. It can be defined as the ratio of the change in drain to source voltage to the change in drain current at constant gate source voltage. It is the reciprocal of the slope of the drain characteristics and is defined as,

$$\begin{aligned} r_d &= \frac{\text{Change in } V_{DS}}{\text{Change in } I_D} \\ r_d &= \left(\frac{\partial V_{DS}}{\partial I_D} \right)_{V_{GS}} = \frac{\Delta V_{DS}}{\Delta I_D} \mid V_{GS} \text{ is held constant} \end{aligned}$$

The reciprocal of r_d is called as drain conductance. It can be denoted by g_d (or) g_m . The typical values of r_d range from $100k\Omega$ to $1m\Omega$ for a JFET and from $10k\Omega$ to $100k\Omega$ for a MOSFET.

3. D.C. Drain Resistance (R_{DS}): This is also called as the static or Ohmic resistance of the channel and is,

$$R_{DS} = \frac{V_{DS}}{I_D}$$

4. Amplification Factor μ : The two parameters g_m and r_d for JFET. It can be defined as the ratio of change in drain to source voltage to the change in gate to source voltage at constant drain current. The third parameter can be known as “amplification factor” is denoted by μ and is defined as,

$$\mu = \frac{\text{change in } V_{DS}}{\text{change in } V_{GS}}$$

$$\mu = - \left(\frac{\partial V_{DS}}{\partial V_{GS}} \right)_{I_D} = \frac{\Delta V_{DS}}{\Delta V_{GS}}, I_D \text{ held constant.}$$

Thus, μ is the ratio of two voltages and it has no limits

For a JFET the value of μ is at the most 100.

Relationship among FET Parameters: As I_D on V_{DS} and V_{GS} , the functional equation can be expressed by,

$$I_D = f(V_{DS}, V_{GS})$$

When the drain voltage is changed by a small amount from V_{DS} to $(V_{DS} + \Delta V_{DS})$ and the gate voltage is changed by a small amount from V_{GS} to $(V_{GS} + \Delta V_{GS})$. The small change in ΔI_D is,

$$\Delta I_D = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \Delta V_{DS} + \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}} \Delta V_{GS} \quad \dots (4)$$

Dividing the both sides of equation (4) with ΔV_{GS} ,

$$\frac{\Delta I_D}{\Delta V_{GS}} = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \cdot \left(\frac{\Delta V_{DS}}{\Delta V_{GS}} \right) + \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}} \cdot \left(\frac{\Delta V_{GS}}{\Delta V_{GS}} \right)$$

$$\frac{\Delta I_D}{\Delta V_{GS}} = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \left(\frac{\Delta V_{DS}}{\Delta V_{GS}} \right) + \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}}$$

If I_D is constant, then $\frac{\Delta I_D}{\Delta V_{GS}} = 0$

$$\text{Hence, } 0 = \left(\frac{\partial I_D}{\partial V_{DS}} \right)_{V_{GS}} \left(\frac{\Delta V_{DS}}{\Delta V_{GS}} \right) + \left(\frac{\partial I_D}{\partial V_{GS}} \right)_{V_{DS}} \Rightarrow 0 = \left(\frac{1}{r_d} \right) (-\mu) + g_m$$

Hence, $\mu = r_d \times g_m \Rightarrow \mu = r_d g_m$

Thus, the parameter μ is the ratio of two similar quantities viz. ratio of two voltages μ is unitless.

Expression for Saturation Drain Current

$$I_{DS} = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

Where I_{DS} = Saturation Drain Current

I_{DSS} = Value of I_{DS} when $V_{GS} = 0$

V_p = Pinch-off voltage

5. Input Resistance (R_i): The ratio of gate- source voltage to drain saturation current $R_i = \frac{V_{GS}}{I_{DSS}}$.

6. Power Deception (P_d): Product of drain- source voltage and drain current $P_d = V_{DS} \times I_D$.

6.8 PINCH-OFF VOLTAGE

Pinch-off Voltage: Channel has Ohmic resistance and there is a potential difference V_{DS} between drain and source, there is a gradual increase in positive potential from source to drain. If the result of this reverse bias at the junctions and hence the width of depletion layers increases as we go from source to drain.

For this reason channel becomes wedge shaped with its narrow end near the drain. If V_{DS} is increased further, the width of channel becomes less and less till it is minimum and two depletion layers meet near the drain end. At this stage the channel is said to be pinched off. The value of V_{DS} at which pinch off takes place is called as pinch off voltage V_p .

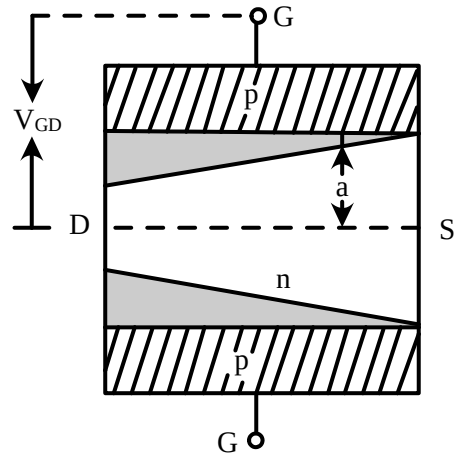


Figure 6.13

Expression for Pinch-off Voltage: Width of the depletion region for a PN junction to which a bias voltage V has been applied is,

$$W = \left[\frac{2\epsilon}{e} (V_B - V) \left(\frac{1}{n_n} + \frac{1}{p_p} \right) \right]^{\frac{1}{2}} \quad \dots (1)$$

But $n_n = N_d$, $p_p = N_a$

The equation (1) becomes,

$$W = \left[\frac{2\epsilon_r \epsilon_0}{e} \left(\frac{N_a + N_d}{N_a N_d} \right) (V_B - V) \right]^{\frac{1}{2}}$$

Where, V_B is contact potential.

N_a is concentration of acceptor ions (number of acceptor ions or holes/cm³ = p_p) on p-side,

N_d concentration of donor ions (number of donor ions or electrons/cm³ = n_n) on n-side.

e is the charge on electron (with negative sign).

ϵ_r is the relative permittivity of material and

ϵ_0 is the permittivity of free space.

Consider V_{GD} is the reverse bias between the gate and drain end of channel. It can be supposed that the gate region is being heavily doped. The depletion layer lies primarily in the n-type channel and its thickness within the p region is neglected. Due to the same reason $N_d \ll N_a$ and therefore N_d can be neglected as compared to N_a .

$$\frac{N_a + N_d}{N_a \times N_d} = \frac{N_a}{N_a N_d} = \frac{1}{N_d}$$

It can be assumed that contact potential V_B is also negligible as compared to V_{GD}

$$V_B - V = -V_{GD}$$

W is the width of depletion layer at the drain end is,

$$W = \left[\frac{2\epsilon_r \epsilon_0}{e} \frac{(-V_{GD})}{N_d} \right]^{\frac{1}{2}}$$

The pinch-off occurs if the value of $-V_{GD} = V_p$ is such that width of depletion layer is equal to channel half width a . Substituting $-V_{GD} = V_p$

[$\therefore$ pinch – off voltage is always a positive quantity] and taking $W=a$. We get,

$$\frac{2\epsilon_r \epsilon_0}{e} \frac{V_p}{N_d} = a^2$$

$$V_p = \frac{ea^2 N_d}{2\epsilon_r \epsilon_0} = \frac{ea^2 N_d}{2\epsilon}$$

For a p-type channel, the pinch off voltage $V_p = \frac{ea^2 N_a}{2\epsilon}$

6.9 APPLICATION OF FET AS VOLTAGE VARIABLE RESISTOR

FET as a Voltage Variable Resistor (VVR)

In most linear applications of field effect transistors the device is operated in the constant current portion of its output characteristics. But some times FET can be operated in the region before pinch-off, where V_{DS} is small. In this region, the FET is useful as a voltage variable resistor i.e., the drain to source resistance is controlled by the bias voltage V_{GS} . In this region the FET is

called as voltage variable resistor (VVR). The variation of r_d with V_{GS} for 2N3277 can be plotted as shown in figure 6.14. The variation of r_d with V_{GS} can be given by the empirical expression is,

$$r_d = \frac{r_0}{1 - KV_{GS}}$$

Where, r_0 = Drain resistance at zero gate bias

K = Constant, dependent on FET type

V_{GS} = Gate to source voltage.

Thus, the FET can be operated in this region acts like a variable passive resistor, so it can be used to vary the voltage gain of a multi-stage amplifier A as the signal level is increased. This action is called as AGC (Automatic Gain Control).

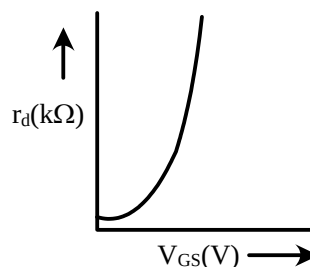


Figure 6.14

6.10 MERITS AND DEMERITS OF FET

Merits of FET

1. It is less noisy
2. It has high input impedance
3. It is thermally stable
4. It is simple to construct
5. It is immune to radiation
6. It is an useful voltage controlled device
7. It can be used as a voltage variable resistor
8. It is compatible for the fabrication in integrated circuits and occupies less space.

Demerits of FET

1. Its gain bandwidth product is small.
2. Its voltage gain is small.

6.11 ADVANTAGES OF JFET

1. A very high input impedance order of 100Ω .
2. Operation of JFET depends on the bulk material current carriers that do not cross junctions.
3. Very high power gain.
4. Negative temperature coefficients.
5. Smaller size longer life and high efficiency.

6. AC drain resistance r_d it is the ratio of change in drain-source voltage to the change in drain current at constant gate source voltage.
7. The amplification factor is the ratio of change in drain source voltage to the change in gate source voltage at constant drain current.
8. Transconductance is the ratio of change in drain current to the change in gate source voltage at constant drain source voltage.

6.12 APPLICATIONS OF JFET

1. JFETs are used in current sources.
2. JFETs are used as voltage controlled resistance devices.
3. JFETs are used in buffer voltage amplifiers.
4. JFETs are used in analog and digital switches.
5. JFETs are used in very high resistance amplifiers.

6.13 FET COMMON SOURCE AMPLIFIER

Common source amplifier is one with no source resistor, so the source is connected to the ground. Common source amplifier is widely used and is similar to CE of a BJT. Circuit configuration as shown in figure 6.15. Common source FET configuration is probably the most widely used of all FET circuit configurations.

Common Source (CS) amplifier is the FET equivalent of the common emitter transistor amplifier configuration. Like the CE amplifier, it is capable of high voltage gain. The CS amplifier has the input applied between the gate and source terminals and the output signal taken across the drain and source terminals. Therefore, the source terminal is common to both the input and output signals. Small signal equivalent circuit using the voltage source model of FET as shown in figure 6.15.

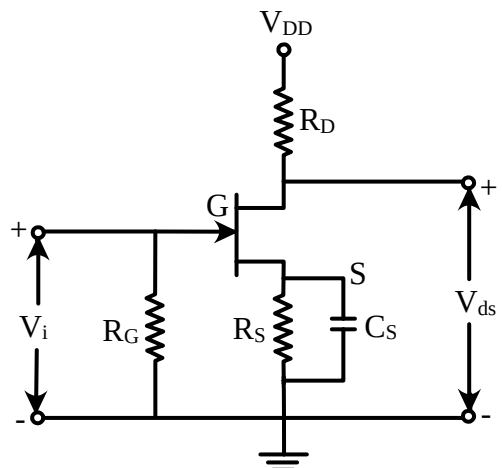


Figure 6.15: Common Source Amplifier

Voltage Gain: The source resistor (R_S) can be used to set the Q point bias by passing CS for mid frequency operation. The output voltage for small signal equivalent circuit is,

$$V_0 = \frac{-R_D}{R_D + r_d} \mu V_{gs}$$

Where $V_{gs} = V_i$, the input voltage.

The voltage gain, $A_V = \frac{V_o}{V_i} = -\frac{\mu R_D}{R_D + r_d}$

Here, negative sign indicates the phase shift of 180° between input voltage and output voltage.

Input Impedance: The input impedance for CS amplifier is, $Z_i = R_G$

For the voltage divider bias as in CE amplifiers of BJT is, $R_G = R_1 \parallel R_2$

Output Impedance: It is the impedance at the output terminals with the input voltage $V_i = 0$.

When $V_i = 0$, $V_{gs} = 0$ and hence $\mu V_{gs} = 0$

The equivalent circuit is as follows for calculating output impedance.

Output impedance is $Z_o = r_d \parallel R_D$

Hence, normally r_d will be far greater than R_D .

Hence $Z_o \approx R_D$.

Note

1. Voltage gain, $A_V = \frac{V_o}{V_i} = -\frac{\mu R_D}{R_D + r_d}$
2. The input impedance is given by, $Z_i = R_G$
3. The output impedance is given by, $Z_o = r_d \parallel R_D \approx R_D$.

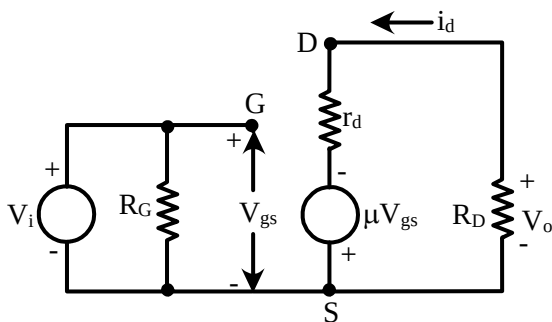


Figure 6.16: Small signal equivalent circuit of CS amplifier

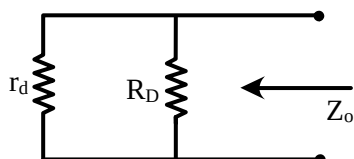


Figure 6.17: Calculation of output impedance

6.14 FET COMMON DRAIN AMPLIFIER

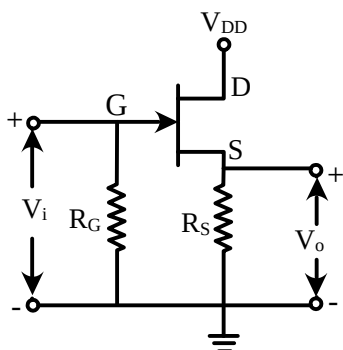


Figure 6.18: Common Drain amplifier

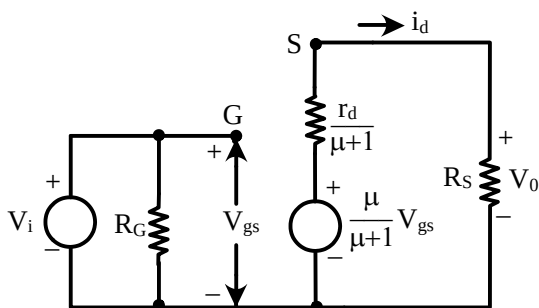


Figure 6.19: Small signal equivalent circuit of CD amplifier

Common Drain Amplifier: Common drain amplifier is similar in configuration to common collector of BJT. This is also referred to source follower. The circuit of such an amplifier is as shown in figure 6.18 and figure 6.19 shows that the small signal equivalent circuit of CD amplifier.

Like the transistor emitter follower, the source follower configuration itself provides a high level of buffering and the high input impedance. Actual input resistance of a FET itself is very high as it is a field effect device. It means the source follower circuit is able to provide an excellent performance as a buffer. Voltage gain is unity, although current gain is high. The input and output signals are in phase. Various parameters of the amplifier can be determined by the following.

Voltage Gain

The voltage gain is, $A_v = \frac{V_o}{V_i}$... (1)

$$\text{Here, } V_o = \frac{R_s}{R_s + \frac{r_d}{\mu + 1}} \times \frac{\mu}{\mu + 1} V_{gd} = \frac{\mu R_s V_{gd}}{(\mu + 1)R_s + r_d}$$

Since, $V_i = V_{gd}$

Substituting the above values in equation (1). We get,

$$\therefore A_v = \frac{V_o}{V_i} = \frac{\frac{\mu R_s V_{gd}}{(\mu + 1)R_s + r_d}}{V_{gd}} = \frac{\mu R_s}{(\mu + 1)R_s + r_d}$$

In CD configuration there is no phase shift between output voltage and input voltage.

Input Impedance: $Z_i = R_G$

Output Resistance: It is measured at the output terminals with an input voltage $V_i = 0$ can be simply calculated from the following equivalent circuit.

As $V_i = 0$, $V_{gd} = 0$, $\frac{\mu}{\mu + 1} V_{gd} = 0$

$$Z_o = \frac{r_d}{\mu + 1} \parallel R_s$$

If $\mu \gg 1$

The voltage gain is slightly less than unity. It implies that an output voltage at the source follows the input voltage at the gate. CD amplifier is thus referred to as source follower and it finds same application as an emitter follower configuration of the bipolar junction transistor.

$$\therefore Z_o = \frac{r_d}{\mu} \parallel R_s = \frac{1}{g_m} \parallel R_s$$

Note

1. Voltage gain,

$$A_v = \frac{\mu R_s}{(\mu + 1)R_s + r_d}$$

2. The input impedance is given by, $Z_i = R_G$

3. The output impedance is given by,

$$Z_o = \frac{r_d}{\mu} \parallel R_s = \frac{1}{g_m} \parallel R_s$$

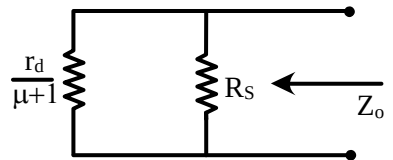


Figure 6.20: Calculation of output impedance

6.15 FET COMMON GATE AMPLIFIER

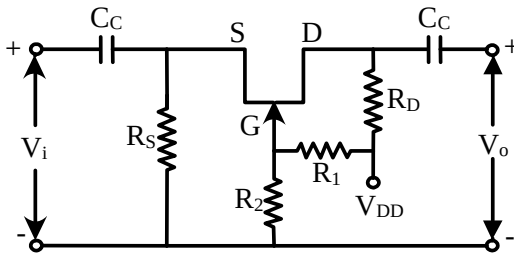


Figure 6.21: Common gate amplifier

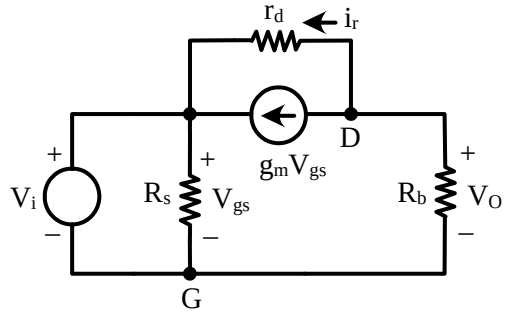


Figure 6.22: Small signal equivalent circuit of CG amplifier

Common Gate Amplifier: FET in common gate configuration can be used as high frequency amplifier. The circuit of such a configuration is as shown in figure 6.21 and figure 6.22 as shows that the small signal equivalent circuit of a CG amplifier. In a CG amplifier, the input signal is applied to the source and the output signal is taken from a drain.

We can replace the FET with its small signal model to determine the various parameters of a CG amplifier. Modified equivalent circuit is as shown in figure 6.23.

Voltage Gain: From figure 6.22, by applying KCL,

$$i_r = i_d - g_m V_{gs}$$

By applying KVL,

$$V_o = (i_d - g_m V_{gs}) r_d + V_{sg}$$

$$V_{sg} = -V_{gs} = V_i \text{ and } i_d = \frac{-V_o}{R_D}$$

$$V_o = \left(\frac{-V_o}{R_D} + g_m V_i \right) r_d + V_i$$

$$A_V = \frac{V_o}{V_i} = \frac{(g_m r_d + 1) R_D}{R_D + r_d}$$

Input Impedance: From figure 6.22 and 6.23

The current through r_d is, $I_{rd} = -I_r = I_1 + g_m V_{gs}$... (1)

But $I_1 = I_{rd} - g_m V_{gs}$... (2)

$$I_{rd} = \frac{V_i - V_o}{r_d} = \frac{V_i - I_{RD} R_D}{r_d}$$

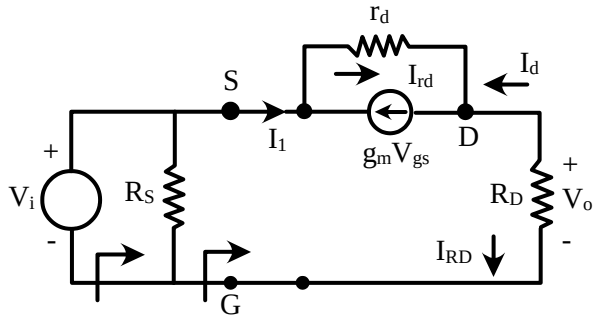


Figure 6.23: Modified equivalent circuit

Substituting above values in equation (2). We get,

$$I_1 = \frac{V_i - I_{RD}R_D}{r_d} - g_m V_{gs}$$

From figure 6.23, Input Voltage is, $V_i = -V_{gs}$

$$I_1 = \frac{V_i - I_{RD}R_D}{r_d} + g_m V_i = \frac{V_i}{r_d} - \frac{I_{RD}R_D}{r_d} + g_m V_i$$

$$I_1 + \frac{I_{RD}R_D}{r_d} = \frac{V_i}{r_d} + g_m V_i \quad \dots (3)$$

$$\frac{I_1 r_d + I_{RD}R_D}{r_d} = \frac{V_i}{r_d} + g_m V_i \quad \dots (4)$$

From figure 6.23, $I_1 = I_{RD}$ in equation (3),

$$\begin{aligned} \frac{I_1 r_d + I_1 R_D}{r_d} &= \frac{V_i}{r_d} + g_m V_i \\ I_1 \left[\frac{r_d + R_D}{r_d} \right] &= V_i \left[\frac{1}{r_d} + g_m \right] \Rightarrow I_1 \left[\frac{r_d + R_D}{r_d} \right] = V_i \left[\frac{1 + r_d g_m}{r_d} \right] \\ \frac{V_i}{I_1} &= \frac{r_d + R_D}{1 + g_m r_d} = Z_i' \end{aligned}$$

Here $Z_i = R_S \parallel Z_i'$

$$Z_i = R_S \parallel \frac{r_d + R_D}{1 + g_m r_d}$$

$g_m r_d \gg 1$ and $r_d \gg R_D$. Then,

$$Z_i = R_S \parallel \frac{r_d}{g_m r_d} = R_S \parallel \frac{1}{g_m}$$

The input impedance is, $Z_i = R_S \parallel \frac{1}{g_m}$

Output Impedance

The output impedance is, $Z_o = r_d \parallel R_D$

$$Z_o \approx R_D \text{ [Since } r_d \gg R_D \text{].}$$

Note

1. Voltage gain,

$$A_V = \frac{(g_m r_d + 1)R_D}{R_D + r_d}$$

2. The input impedance is given by, $Z_i = R_S \parallel \frac{1}{g_m}$

3. The output impedance is given by, $Z_o \approx R_D$.

6.16 COMMON SOURCE JFET AMPLIFIERS

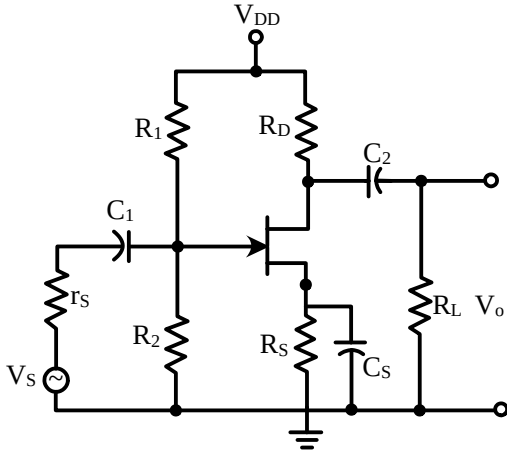


Figure 6.24: Common Source JFET amplifiers

Figure 6.24 shows a JFET CS amplifier arrangement. Here, an resistors R_1, R_2 and R_S provides the stable operating point over a wide range of a temperature and the transfer characteristics.

Capacitors C_1, C_2 and C_S can be assumed to have the reactance's in a operating frequency range that are small compared with $R_1 \parallel R_2, R_L$ and R_S .

The equivalent circuit for JFET CS amplifier as shown in figure 6.25.

From the equivalent circuit for JFET CS amplifier,

The output voltage is,

$$V_o = -g_m V_{gs} (r_o \parallel R_D \parallel R_L) = -g_m V_i (r_o \parallel R_D \parallel R_L) \quad \dots (1)$$

Voltage Gain: The voltage gain is,

$$A_V = \frac{V_o}{V_i} = -g_m (r_o \parallel R_D \parallel R_L) \quad \dots (2)$$

The voltage gain, source resistance into consideration as,

$$A_{VS} = \frac{V_o}{V_S} = \frac{R_B}{r_s + R_B} \times A_V = \left(\frac{R_B}{R_B + r_s} \right) g_m (r_o \parallel R_D \parallel R_L) \quad \dots (3)$$

In FET amplifiers R_B is very large and r_o is also high.

$$A_V = A_{VS} = -g_m (R_D \parallel R_L) \quad \dots (4)$$

Input Impedance (Z_i)

The input impedance is,

$$Z_i = \frac{V_i}{I_i} = R_B = R_1 \parallel R_2 \quad \dots (5)$$

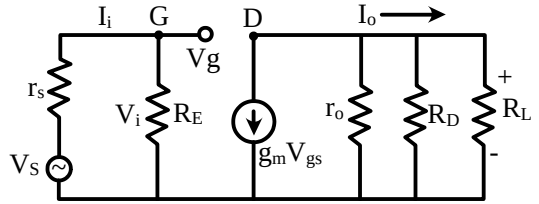


Figure 6.25: Equivalent circuit for CS JFET amplifier

Current Gain A_i

$$A_i = \frac{I_o}{I_i} = \frac{V_o / R_L}{V_i / Z_L} = A_v \cdot \frac{Z_i}{R_L} \quad \dots (6)$$

Output Impedance Z_o

$$Z_o = r_o \parallel R_D \parallel R_L \quad \dots (7)$$

6.17 COMPARISON OF CS, CD AND CG FET AMPLIFIERS

CS Amplifiers	CD Amplifiers	CG Amplifiers
Voltage gain is, $A_v = \frac{V_o}{V_i} = -\frac{\mu R_D}{R_D + r_d}$	Voltage gain is, $A_v = \frac{\mu R_S}{(\mu + 1)R_S + r_d}$	Voltage gain is, $A_v = \frac{(g_m r_d + 1)R_D}{R_D + r_d}$
Input impedance is, $Z_i = R_G$	Input impedance is, $Z_i = R_G$	Input impedance is, $Z_i = R_S \parallel \frac{1}{g_m}$
Output impedance is, $Z_o = r_d \parallel R_D \approx R_D$	Output impedance is, $Z_o = \frac{r_d}{\mu} \parallel R_S = \frac{1}{g_m} \parallel R_S$	Output impedance is, $Z_o \approx R_D$
Phaseshift of CS amplifier is 180° .	Phaseshift of CD amplifier is 0° .	Phaseshift of CD amplifier is 0° .
Gain of CS amplifier is generally greater than one.	Gain of CD amplifier is slightly less than one.	Gain of CG amplifier offers gain of more than one.
Output resistance of CS can be determined by R_D .	Output resistance of CD is low $\approx 100\Omega$.	Output resistance of CC can be determined by R_D .
It can be used as a invert input.	It can be used as a voltage buffer.	It can be used as a current buffer.

6.18 SOLVED PROBLEMS

Example-1: A common source FET amplifier circuit shown in figure 6.26 with unbypassed

R_S has the following circuit parameters $R_d = 15k\Omega$, $R_S = 0.5k\Omega$, $R_g = 1M\Omega$, $g_m = 5m \text{ mho}$ and $V_{DD} = 20 \text{ V}$. Calculated amplification parameter A_v and R_o .

Sol.

Given data,

For a common source FET amplifier,

$$\begin{array}{llll} R_d = 15k\Omega, & R_S = 0.5k\Omega, & R_g = 1M\Omega \text{ and} & r_d = 5k\Omega \\ g_m = 5m \text{ mho}, & A_v = ?, & R_o = ? & \end{array}$$

We know,

1. Amplification Factor

$$\mu = r_d g_m = 5 \times 10^3 \times 5 \times 10^{-3} = 25$$

2. Voltage Gain

$$A_V = \frac{V_o}{V_i} = -\frac{\mu R_D}{R_D + r_d} = -\frac{25 \times 15 \times 10^3}{15 \times 10^3 + 5 \times 10^3}$$

$$A_V = -\frac{375 \times 10^3}{20 \times 10^3} = -18.75$$

3. Output Resistance

Output resistance is, $R_o = R_d \parallel r_d$

$$R_o = \frac{R_d \times r_d}{R_d + r_d} = \frac{15 \times 10^3 \times 5 \times 10^3}{15 \times 10^3 + 5 \times 10^3} = \frac{75 \times 10^6}{20 \times 10^3}$$

$$\therefore R_o = 3.75 \text{ k}\Omega$$

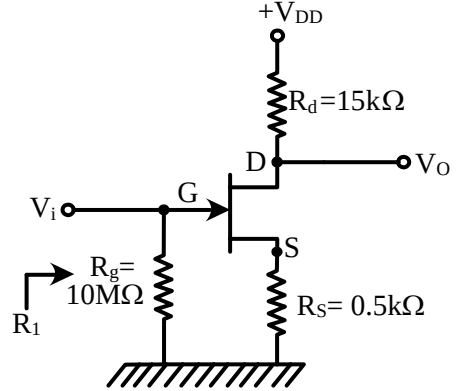


Figure 6.26

Example-2: In the CD amplifier circuit, let $R_s = 4 \text{ k}\Omega$, $R_G = 10 \text{ M}\Omega$, $\mu = 50$ and $r_d = 35 \text{ k}\Omega$.

Evaluate the voltage gain A_V , input impedance Z_i and output impedance Z_o .

Sol.

Given data,

$$R_s = 4 \text{ k}\Omega, \quad R_G = 10 \text{ M}\Omega, \quad \mu = 50 \text{ and } r_d = 35 \text{ k}\Omega$$

1. Voltage Gain

The voltage gain,

$$A_V = \frac{V_o}{V_i} = \frac{\mu R_s}{(\mu + 1)R_s + r_d} \quad \dots (1)$$

Substituting the given values in equation (1). We get,

$$A_V = \frac{50 \times 4 \times 10^3}{(50 + 1) \times 4 \times 10^3 + 35 \times 10^3} = 0.0836$$

The positive value indicates that V_o and V_i are in phase and further note that $A_V < 1$ for CD amplifier.

2. Input Impedance

The input impedance is, $Z_i = R_G = 10 \text{ M}\Omega$

3. Output Impedance

$$\text{Output impedance, } Z_o = \left(\frac{1}{g_m} \right) \parallel R_s = \left(\frac{r_d}{\mu} \right) \parallel R_s$$

$$Z_o = \frac{35 \times 10^3}{50} \parallel 4 \times 10^3 = 595.7 \Omega$$

Example-3: In the CG amplifier, let $R_D = 3k\Omega$, $R_S = 2k\Omega$, $g_m = 1.43 \times 10^{-3} \text{ mho}$ and $r_d = 20k\Omega$. Evaluate Voltage gain, A_v , input impedance Z_i and output impedance Z_o .

Sol.

Given data,

$$R_D = 3k\Omega, \quad R_S = 2k\Omega, \quad g_m = 1.43 \times 10^{-3} \text{ mho} \quad \text{and} \quad r_d = 20k\Omega$$

1. Voltage Gain A_v

The voltage gain is, $A_v = \frac{(g_m r_d + 1)R_D}{R_D + r_d}$

$$A_v = \frac{(1.43 \times 10^{-3} \times 20 \times 10^3 + 1)3 \times 10^3}{(3 \times 10^3 + 20 \times 10^3)} = \frac{(1.43 \times 20 + 1)3 \times 10^3}{23 \times 10^3} = 3.86$$

2. Input Impedance: The input impedance is,

$$Z_i = R_S \parallel \frac{1}{g_m}$$

$$\begin{aligned} 2 \times 10^3 \parallel \frac{1}{1.43 \times 10^{-3}} &= 2 \times 10^3 \parallel \frac{10^3}{1.43} \\ &= \frac{2 \times 10^3 \times \frac{10^3}{1.43}}{2 \times 10^3 + \frac{10^3}{1.43}} = \frac{2 \times 10^3 \times 10^3}{2 \times 10^3 \times 1.43 + 10^3} = 0.5k\Omega \end{aligned}$$

3. Output Impedance

The output impedance is, $Z_o \approx R_D \approx 3k\Omega$

Example-4: A JFET has $I_{DSS} = 6\text{mA}$ and $V_{GS(off)} = -2.5\text{V}$. Calculate drain current at $V_{GS} = -1\text{V}$.

Sol.

Given data,

$$I_{DSS} = 6\text{mA}, \quad V_{GS(off)} = -2.5\text{V} \quad \text{and} \quad V_{GS} = -1\text{V}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad \dots (1)$$

Substituting the given values in equation (1). We get,

$$\therefore I_D = 6 \times 10^{-3} \left[1 - \frac{1}{2.5} \right]^2 = 6 \times 10^{-3} \times 0.36 = 2.16\text{mA}$$

Example-5: The drain current of JEET shown in figure 6.27 is given by

$$I_D = 10 \left(1 + \frac{V_{GS}}{4} \right)^2 \text{ mA}.$$

Calculate the quiescent values of I_{DS} , V_{DS} and V_{GS} .

Sol.

Given equation is,

$$I_D = 10 \left(1 + \frac{V_{GS}}{4} \right)^2 \text{ mA} \quad \dots (1)$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2 \quad \dots (2)$$

Comparing the equation (1) with the standard form equation (2), we get

$$I_{DSS} = 10 \text{ mA}; \quad V_p = -4 \text{ V}$$

Where V_{GS} = Voltage drop across

$$R_S = -I_D \times 500$$

Substituting the V_{GS} in the given equation (1). We get,

$$I_D = 10 \left[1 - \frac{500 I_D}{4} \right]^2$$

$$I_D = 10 - 2.5 I_D + 0.156 I_D^2 \quad \dots (3)$$

By solving equation (3). We get,

$$I_D = 19.00 \text{ mA (or) } 3.36 \text{ mA}$$

The first value is absurd as it leads to

$$V_D = I_D (R_D - I R_S) = 104 \text{ V}$$

$$\therefore I_D = 3.36 \text{ mA}$$

$$\therefore V_{GS} = -I_D \times 0.5 \text{ k} = 1.68 \text{ V}$$

$$\therefore V_{GS} = 30 - 3.36 \times 10^{-3} (5 \text{ k} + 0.5 \text{ k}) = 11.52 \text{ V}$$

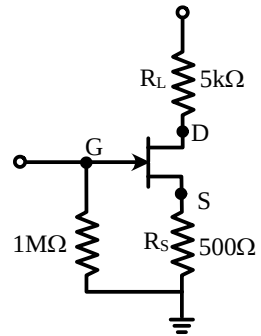


Figure 6.27

Example-6: For a P-channel JFET the physical width of the channel is $2 \times 10^{-1} \text{ cm}$ and channel resistivity is $20 \Omega \text{ cm}$. Determine the pinchoff voltage.

Sol.

Given data, $2a = 2 \times 10^{-1} \text{ cm} \Rightarrow a = 1 \times 10^{-1} \text{ cm}$

Now $\sigma = N_D q \mu_n$

$$\rho = \frac{1}{N_D q \mu_n}$$

$$N_D = \frac{1}{20 \times 1.6 \times 10^{-19} \times 3800} = 8.23 \times 10^{13}$$

$$V_p = \frac{q N_D}{2 \epsilon} a^2$$

$$\therefore V_p = \frac{1.6 \times 10^{-19} \times 8.23 \times 10^{13}}{2 \times 12 \times 8.85 \times 10^{-12}} \times 10^{-8} = 6.12 \text{ V}$$

Example-7: Calculate the dynamic resistance of a JFET having amplification factor 80 and transconductance $200\mu\text{mho}$.

Sol.

Amplification factor $\mu = 80$

Transconductance $g_m = 200 \times 10^{-6} \text{ mho}$

Dynamic resistance $r_d = ?$

$$\mu = g_m \times r_d$$

$$\therefore r_d = \frac{\mu}{g_m} = \frac{80}{200 \times 10^{-6}} = 400 \times 10^3 \Omega = 400 \text{ k}\Omega$$

Example-8: Determine the pinch-off voltage for n-channel silicon FET with a channel width of $5.6 \times 10^{-4} \text{ cm}$ and a donor concentration of 10^{15} cm^{-3} . Given that dielectric constant for Si is 12.

Sol.

We know,

$$V_p = \frac{eN_D}{2\epsilon} a^2$$

Here $2a = 5.6 \times 10^{-4} \text{ cm} = 5.6 \times 10^{-6} \text{ m}$

$$\epsilon = 12\epsilon_0 = 12 \times 8.86 \times 10^{-12} \text{ F/m}$$

$$N_D = 10^{15} \text{ cm}^{-3} = 10^{21} \text{ m}^{-3}$$

Since $e = 1.6 \times 10^{-19} \text{ coulomb}$

$$V_p = \frac{1.6 \times 10^{-19} \times 10^{21}}{2 \times 12 \times 8.86 \times 10^{-12}} \times (2.8 \times 10^{-6})^2 = 5.9 \text{ V}$$

Example-9: N-Channel JFET has a pinch-off voltage off -4.5 V and $I_{DSS} = 9 \text{ mA}$. At what value of V_{GS} in the pinch-off region will I_D be equal to 3 mA ?

Sol.

The Shockley equation is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

$$3 = 9 \left(1 - \frac{V_{GS}}{-4.5} \right)^2 \Rightarrow \frac{3}{9} = \left(1 + \frac{V_{GS}}{4.5} \right)^2$$

$$1 + \frac{V_{GS}}{4.5} = \frac{1.732}{3} = 0.577$$

$$\frac{V_{GS}}{4.5} = 0.577 - 1 = -0.423$$

$$V_{GS} = -0.423 \times 4.5 = -1.9 \text{ V}$$

Example-10: In a field effect transistor when value of gate voltage is changed from (-3.0) V to (-2.9)V, the drain current increases for 1mA to 1.2mA. Determine the mutual conductance of the transistor.

Sol.

The mutual conductance

$$\therefore g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{(1.2-1)\text{mA}}{-2.9-(-3)\text{V}} = 2\text{mS}.$$

Check your Knowledge

1. Describe the construction and working of N-channel FET.

Ans. _____

2. Define drain resistance (r_d), transconductance (g_m), amplification factor (μ) and obtain relation among them.

Ans. _____

3. Comparison of characteristics of FET amplifiers.

Ans. _____

4. Discuss small signal model of FET and its parameters.

Ans. _____

5. Explain the output and transfer characteristics of JFET. Explain how a JFET is superior to BJT?

Ans. _____

6. Define FET parameters and give the relationship among them.

Ans. _____

7. Draw the schematic representation of N-channel and P-channel FETs.

Ans. _____

8. Explain the construction of JFET. Describe its drain characteristics showing Ohmic region and pinch-off region.

Ans. _____

9. Give the working mechanism of a FET.

Ans. _____

10. Describe the operation of FET and draw its output characteristics.

Ans. _____

6.19 FET Biasing

The biasing of a field effect transistor is not as critical as of biasing a bipolar junction transistor since the operating point selection in a FET is base only on three parameters, namely, I_D , V_{GS} , V_{DS} whereas for a BJT the dimensions of operating point should be selected in the constant current region.

It is necessary to maintain the operating point Q stable in the central portion of the pinch voltage region in FET biasing. The Q-point should be independent of device parameter variations and an ambient temperature changes. It can be achieved by suitable selecting the gate to source voltage and drain current which is referred to as biasing.

Biasing an FET amplifier circuit is similar to BJT amplifiers. We will use components external to transistor and dc sources to define a predictable and stable operating point, about which the circuit may provide linear amplification. Bias stability in FET amplifiers means that the dc drain current (I_D) stays as constant as possible with variations in operating conditions and device parameters. The different biasing techniques are,

1. Fixed bias method
2. Self -bias method
3. Voltage divider bias method

The purpose of all the four biasing techniques is to provide a stable Q point in the constant current or saturation region of JFET's output characteristic against widespread variations in parameters of a JFET due to the fabrication variations and temperature fluctuations.

The variation of 50% in the parameters of JFET is not unusual. To ensure that Q-point is in the constant current or saturation region. The care must be taken to see that quiescent value of $|V_{DS}|$. The drain-to-source voltage must be greater than $|V_p| - |V_{GS}|$, where V_{GS} is gate-to-source voltage and region by the JFET's equation of the form is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

The I_{DSS} is drain current with the gate source terminal shorted.

6.19.1 Fixed-Bias Configuration

The FET device needs the DC bias to setting of gate-source voltage V_{GS} to give the desired drain current I_D . For the JFET drain current is limited by the saturation current I_{DSS} . Thus, the FET has such a high input impedance and it does not allow the gate current to flow and the dc voltage of the gate set by a voltage divider or the fixed battery voltage is not affected or loaded by FET.

The fixed dc bias can be obtained by using the battery V_{GG} . This battery can be ensures that the gate is always negative with respect to source and no current flows through resistor R_G and gate terminal i.e., $I_G = 0$.

The battery provides a voltage V_{GS} to bias the N-channel JFET but no resulting current can be drawn from the battery V_{GG} . The Resistor R_G is included to allow any ac signal applied through capacitor C to develop across R_G . Any ac signal will develop across R_G and the dc voltage drop across R_G is equal to $I_G R_G$ i.e., 0 volt.

The gate-source voltage V_{GS} is,

$$V_{GS} = -V_G - V_S = -V_{GG} - 0 = -V_{GG} \quad \dots (1)$$

The drain-source current I_D is then fixed by the gate-source voltage as determined by equation. This current causes a voltage drop across the drain resistor R_D and is,

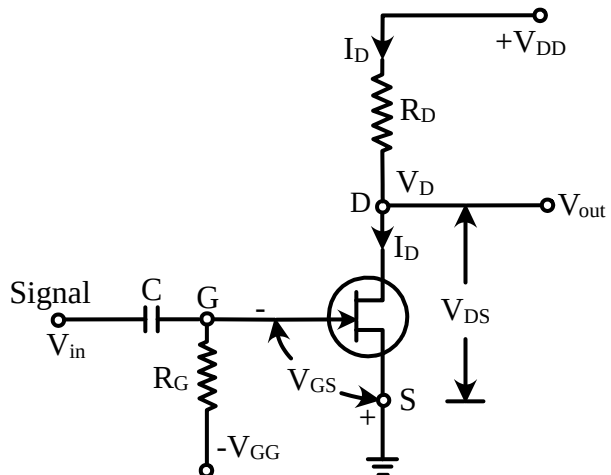


Figure 6.28: Fixed biasing circuit for JFET

$$V_{DD} = I_D R_D + V_{DS} \quad \dots (2)$$

$$V_{DS} = V_{DD} - I_D R_D$$

$$I_D = \frac{V_{DD} - V_{DS}}{R_D}$$

We know that JFET always operates in depletion mode i.e., ' $V_{GG} = -V_e$ '

By applying KVL at input circuit, we get $V_{GG} = -V_{GS}$

By applying KVL at output circuit, we get $V_{DS} = V_{DD} - I_D R_D$

6.19.2 Self-Bias Configuration

The basic circuit of the N-channel FET self biased configuration has been shown in figure 6.29. This is the most common method for biasing a JFET. Hence, no gate current flows through the reverse-biased gate-source, the gate current $I_G = 0$. Therefore,

$$V_G = I_G R_G = 0V$$

With a drain current I_D and at the voltage source S is,

$$V_S = I_D R_S \quad \dots (1)$$

The drain voltage is $V_D = V_{DD} - I_D R_D$

The gate-source voltage is, $V_{GS} = V_G - V_S = 0 - I_D R_S = -I_D R_S \quad \dots (2)$

$$V_{DD} = I_D R_D + V_{DS} + I_D R_S$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S) \quad \dots (3)$$

$$I_D [R_D + R_S] = V_{DD} - V_{DS}$$

$$I_D = \frac{V_{DD} - V_{DS}}{R_D + R_S}$$

$$\text{If } I_D = 0 \quad V_{DD} = V_{DS}$$

$$\text{If } V_{DS} = 0 \Rightarrow I_D = \frac{V_{DD}}{R_D + R_S} \quad \dots (4)$$

The voltage V_{GS} and no external source is required for biasing and this is the reason it is called as self-biasing. DC conditions of JFET amplifier is fully specified. Self biasing of a JFET stabilizes its quiescent operating point against any change in its parameters like transconductance.

Consider the given JFET can be replaced by another JFET having the double conductance then drain current will also try to be double but since any increase in voltage drop across R_S . Hence, the gate-source voltage V_{GS} becomes more negative and thus increase in drain current can be reduced.

From equation (4), I_D increase then the voltage drop across the resistor increases.

- If I_D is current is high then the drop across, the resistor R_S is also very high.

$$I_S R_S = V_S = I_D R_S \uparrow$$

Therefore V_{GG} also increases negatively ($V_{GG} = -V_S$)

The channel width decreases, finally the drain current decreases. Suppose if the drain current is very low then the drop across the resistance R_S is also very low. Therefore V_{GG} is also very low.

6.19.3 Voltage Divider Bias

Figure 6.29 (a) shows the voltage divider bias circuit and its Thevenin's equivalent circuit as shown in figure 6.29(b). Resistors R_1 and R_2 can be connected on gate side forms the voltage divider.

The gate voltage is,

$$V_{GG} = \left(\frac{R_2}{R_1 + R_2} \right) V_{DD} \text{ and } R_g = \frac{R_1 R_2}{R_1 + R_2}$$

The circuit can be reduced.

The bias line satisfies the equation is,

$$V_{GS} = V_{GG} - I_D R_S$$

Drain to ground voltage is, $V_D = V_{DD} - I_D R_D$

$$\begin{aligned} V_{DD} &= I_D R_D + V_{DS} + I_S R_S \\ &= I_D R_D + V_{DS} + I_D R_S \end{aligned}$$

$$V_{DD} = I_D (R_D + R_S) + V_{DS}$$

If the gate voltage V_{GG} is very large as compared to gate to source voltage V_{GS} and the drain current is approximately constant. The voltage divider bias is less effective with the JFET than BJT.

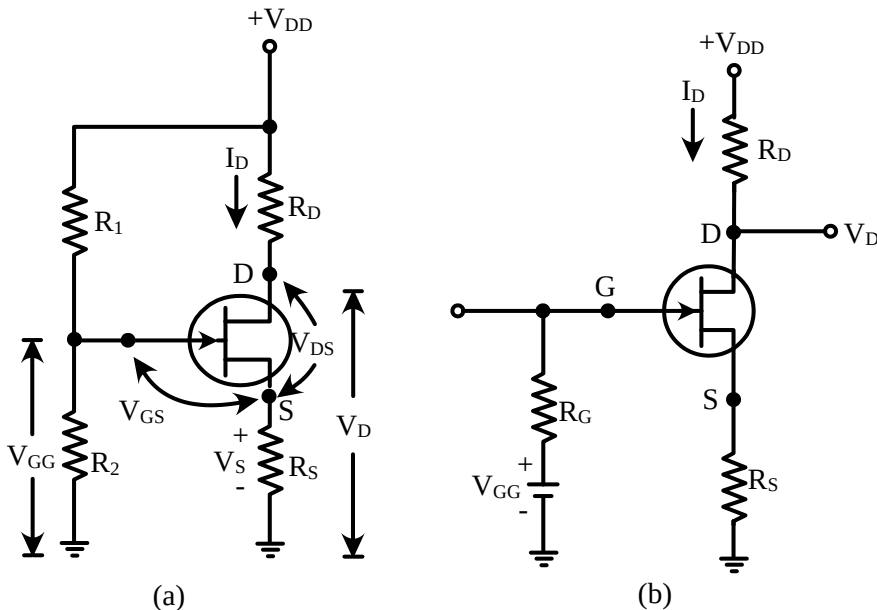


Figure 6.29: (a) Voltage divider bias circuit and (b) Thevenin's equivalent circuit

This is because in BJT, $V_{BE} \approx 0.7V$ (Si) with only minor variations from one transistor to another. But in JFET, V_{GS} can vary several volts from one JFET to another.

6.19.4 Potential Divider Biasing

The slightly modified form of dc bias is provided by the circuit as shown in figure 6.30. The resistors R_{G1} and R_{G2} form the potential divider across drain supply V_{DD} . The voltage V_2 across R_{G2} provides the necessary bias as shown in figure 6.30. The additional gate resistor R_{G1} from gate to supply voltage can be facilitates in larger adjustment of dc bias point and permits use of larger valued R_S .

The gate is reverse biased so that $I_G = 0$ and gate voltage is,

$$V_G = V_2 = (V_{DD} / R_{G1} + R_{G2}) \times R_{G2}$$

$$V_{GS} = V_G - V_S = V_G - I_D R_S \quad \dots (1)$$

The circuit is so designed that $I_D R_S$ is greater than V_G so that V_{GS} is negative. It can be provides correct bias voltage. The operating point can be determined as,

$$I_D = (V_2 - V_{GS}) / R_S$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S) \quad \dots (2)$$

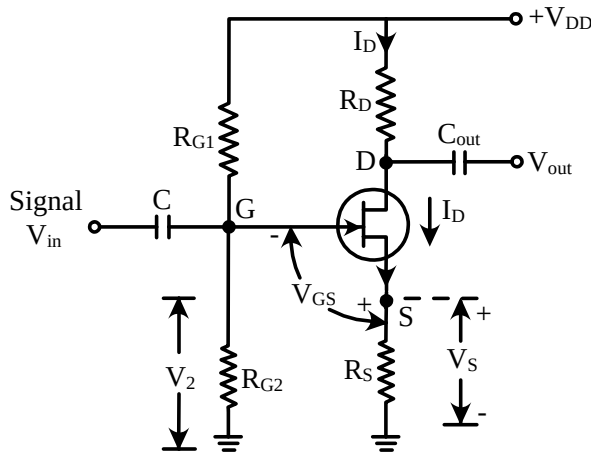


Figure 6.30: Potential Divider Bias Circuit for N-Channel JFET

Check your Knowledge

1. Explain in detail about different biasing techniques.

Ans. _____

2. Write in detail about voltage divider bias and potential divider bias.

Ans. _____

6.20 OUTCOMES

- Able to design a JFET based amplifier for practical applications.
- Ability to perform analysis on various configuration of JFET.
- Select a suitable biasing method for the JFET stability.

6.21 SUMMARY

The FETs are unipolar devices because they operate only with one type of charge carrier. P-channel JFET's channel is doped with acceptor impurities meaning that the flow of current through the channel is positive (hence the term P-channel) in the form of holes. N-channel JFET's have a greater channel conductivity (lower resistance) than their equivalent P-channel types. Since electrons have a higher mobility through a conductor compared to holes. The input impedance of FET is more than that of BJT because the input circuit of FET is reverse bias whereas the input circuit of BJT is forward bias.

FET amplifiers have certain advantages over BJT amplifiers like high input impedance. However, the BJT normally has a higher voltage gain. These are also similarities in the three amplifier configurations of FETs and BJTs. Common-drain has no phase shift, the gain slightly less than 1 and the output is taken from the source. Common-gate has no phase shift and low input resistance. FET amplifier configuration is similar to BJT amplifiers. Trans-conductance (g_m) relates the drain current (ac output) to the ac input voltage (V_{gs}). The input resistance for a FET at the gate is extremely high. Common-source is the most used type of a FET amplifier and has a phase inversion is 180° .

6.22 REVIEW QUESTIONS

I. Short Answer Questions

1. Distinguish between BJT and FET.
2. Discuss the advantages of FET over BJT.
3. Write the advantages and disadvantages of JFET over BJT.
4. In what respect does a FET differ from a BJT?
5. Give the working mechanism of a FET.
6. Explain the differences between biasing a BJT and a FET.
7. Explain the methods used for biasing JFET.

8. Define pinch-off voltage of a JFET. Sketch the depletion region before and after pinch off.
9. Describe the operation of FET and draw its output characteristics.
10. Sketch small signal FET model.
11. Describe the construction and working of N-channel FET.
12. Describe construction and working of JFET.
13. Define drain resistance (r_d), transconductance (g_m), amplification factor (μ) and obtain relation among them.
14. Define the amplification factor.
15. Comparison of performance quantities of the FET amplifiers.
16. Comparison of characteristics of FET amplifiers.

II. Long Answer Questions

1. Explain the construction, working and V-I characteristics of JFET.
2. Draw the schematic representation of N-channel and P-channel FETs.
3. Explain with a neat circuit diagram, an experiment to determine the output and transfer characteristics of a FET.
4. Explain the construction and working of a FET. How do you determine its characteristics?
5. Explain the construction and characteristics of a JFET. Give its small signal model and explain the different terms used.
6. Mention three important differences between a transistor and a FET as circuit element. Give the experimental set up for FET characteristics and discuss them.
7. Discuss the output and transfer characteristics of FET with experimental set-up.
8. Discuss small signal model of a FET. Draw its small signal equivalent circuit.
9. Discuss small signal model of FET and its parameters.
10. Explain the terms, pinch-off voltage, ON-resistance and pinch-off region.
11. Define FET parameters and give the relationship among them.
12. Describe the experimental set up to draw the output and transfer characteristics of a JFET. Explain how FET parameters are determined from these characteristics.
13. Describe the experimental setup to draw the output and transfer characteristics of JFET. Explain how FET parameters can be determined from these characteristics?
14. Explain the output and transfer characteristics of JFET. Explain how a JFET is superior to BJT?
15. Describe the construction and working of JFET with a neat diagram. Explain its various biasing modes.

16. Explain experimental arrangement for studying output and transfer characteristics and to determine FET parameters.
17. Explain the construction of JFET. Describe its drain characteristics showing Ohmic region and pinch-off region.
18. Derive the expressions for FET as a common source amplifiers.
19. Derive the expressions for FET as a common drain amplifiers.
20. Derive the expressions for FET as a common gate amplifiers.
21. Explain in detail about generalized FET Amplifiers.
22. Derive the expressions for JFET as a common source amplifiers.

III. Exercises

1. Determine the value of R_D and R_S for a self biased P-channel JFET using the following parameters $V_P = 5V$, $I_{DSS} = 12mA$, $V_{DD} = 12V$, $I_D = 5mA$ and $V_{DS} = 6V$.
[Ans : $R_P = 1.5\Omega$, $R_S = 525k\Omega$]
2. Determine the values of R_S required in self biased circuit of N-channel JFET with $I_{DSS} = 50mA$, $V_P = -10V$, $V_{GSQ} = -5V$.
[Ans : 400Ω]
3. Design a self-bias configuration for a n-channel JFET having $I_{DSS} = 8mA$ and $V_P = -6V$. The Q-point is at $I_{DQ} = 4mA$ with a supply voltage of 14V. Assume $R_D = 3R_S$.
[Ans : $R_S = 0.43k\Omega$, $R_D = 1.3k\Omega$]

6.23 MULTIPLE CHOICE QUESTIONS

1. The JFET is a ()
(a) Voltage Controlled device (b) Current Controlled device
(c) Voltage and current Controlled device (d) Regulator device
2. The input resistance of a FET is ____ ()
(a) Nearly zero (b) Nearly one (c) Nearly infinite (d) Nearly two
3. The gate-source junction of a JFET should be ()
(a) Forward biased (b) Reverse biased
(c) Either forward or reverse biased (d) In saturation
4. The pinch-off voltage has same magnitude as the ()
(a) Gate Voltage (b) Drain source voltage
(c) Gate source voltage (d) Gate source cut-off voltage

5. For small values of drain to source voltage V_{DS} , a JFET behave like a ()
(a) Resistor (b) Constant current
(c) Constant voltage source (d) None of the above
6. When the gate voltage becomes more negative in n-channel JFET, the channel between the depletion layers ()
(a) Shrinks (b) More conducts (c) Expands (d) None
7. An operational amplifier has ()
(a) Zero input resistance (b) Infinite input resistance
(c) Variable input resistance (d) Based on input
8. The voltage follower has a ()
(a) Unity gain (b) Infinite gain (c) Zero gain (d) None
9. In a P-channel JFET, the charge carriers are ____ ()
(a) Electrons (b) Holes
(c) Both electrons and holes (d) Either electrons or holes
10. If the reverse bias voltage is increased then the width of the conducting channel of JFET is ()
(a) Increased (b) Decreased (c) Remains same (d) None
11. The input control parameter if JFET is ()
(a) Source voltage (b) Gate voltage
(c) Drain voltage (d) Channel
12. The constant current region of a JFET lies between ()
(a) Cut-off and saturation (b) Cut-off and pinch-off
(c) Pinch off and break down regions (d) 0 and I_{DSS}
13. The trans-conductance of a JFET ranges from ()
(a) 0.5 to 30mA/V (b) 100 to 500mA/V
(c) 500 to 750mA/V (d) Above 750mA/V
14. FETs have similar properties to ()
(a) PNP transistors (b) NPN transistors
(c) Thermionic valves (d) Unijunction transistors
15. For small values of drain-to-source voltage, JFET behaves like a ()
(a) Resistor (b) Constant-current source

- (c) Constant-voltage source (d) Negative resistance
16. In a JFET, the primary control on drain current is exerted by ()
(a) Channel resistance (b) Size of depletion regions
(c) Voltage drop across channel (d) Gate reverse bias
17. In a JFET, as external bias applied to the gate is increased ()
(a) Channel resistance is increased
(b) Drain current is increased
(c) Pinch-off voltage is reached at lower values of I_D
(d) Size of depletion regions is reduced.
18. The transconductance ' g_m ' of a JFET is equal to ()
(a) $-\frac{2I_{DSC}}{V_P}$ (b) $\frac{2}{|V_P|} \sqrt{I_{DSS} I_D}$
(c) $-\frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right)$ (d) $\frac{I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right)$
19. The extremely high input impedance of a MOSFET is primarily due to the ()
(a) Absence of its channel
(b) Negative gate-source voltage
(c) Depletion of current carriers
(d) Extremely small leakage current of its gate capacitor
20. Threshold voltage of an N-channel enhancement mode MOSFET is 0.5 V, when the device is biased at a gate voltage of 3 V, pinch-off would occur at a drain voltage of ()
(a) 1.5 V (b) 2.5 V (c) 3.5 V (d) 4.5 V

Answers

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (c) | 3. (b) | 4. (d) | 5. (a) | 6. (a) |
| 7. (c) | 8. (a) | 9. (b) | 10. (b) | 11. (b) | 12. (c) |
| 13. (a) | 14. (b) | 15. (a) | 16. (d) | 17. (c) | 18. (c) |
| 19. (d) | 20. (c) | | | | |